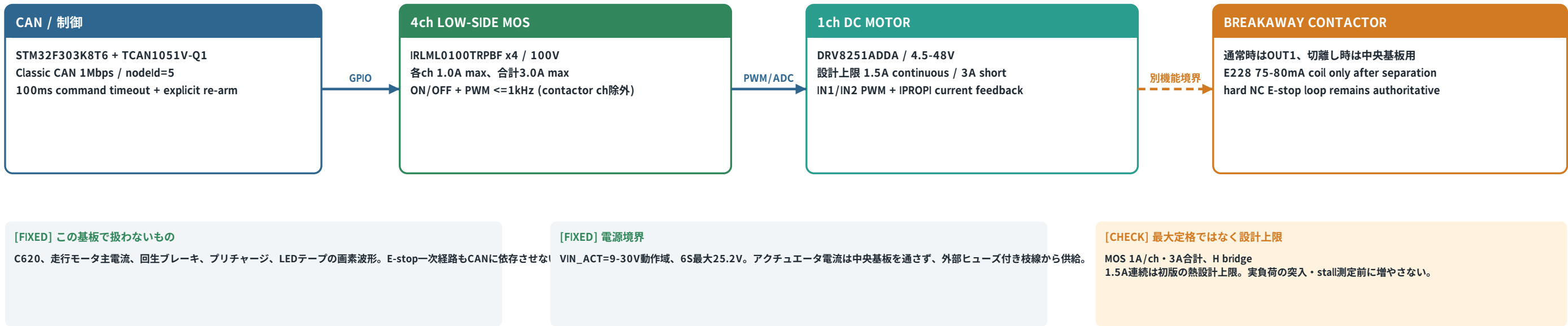


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12-24V補機をCANで安全にON/OFF・PWM制御するローカルノード

中央Teensyは高レベル指令のみ。負荷電流・PWM・タイムアウト処理はノード内で完結。

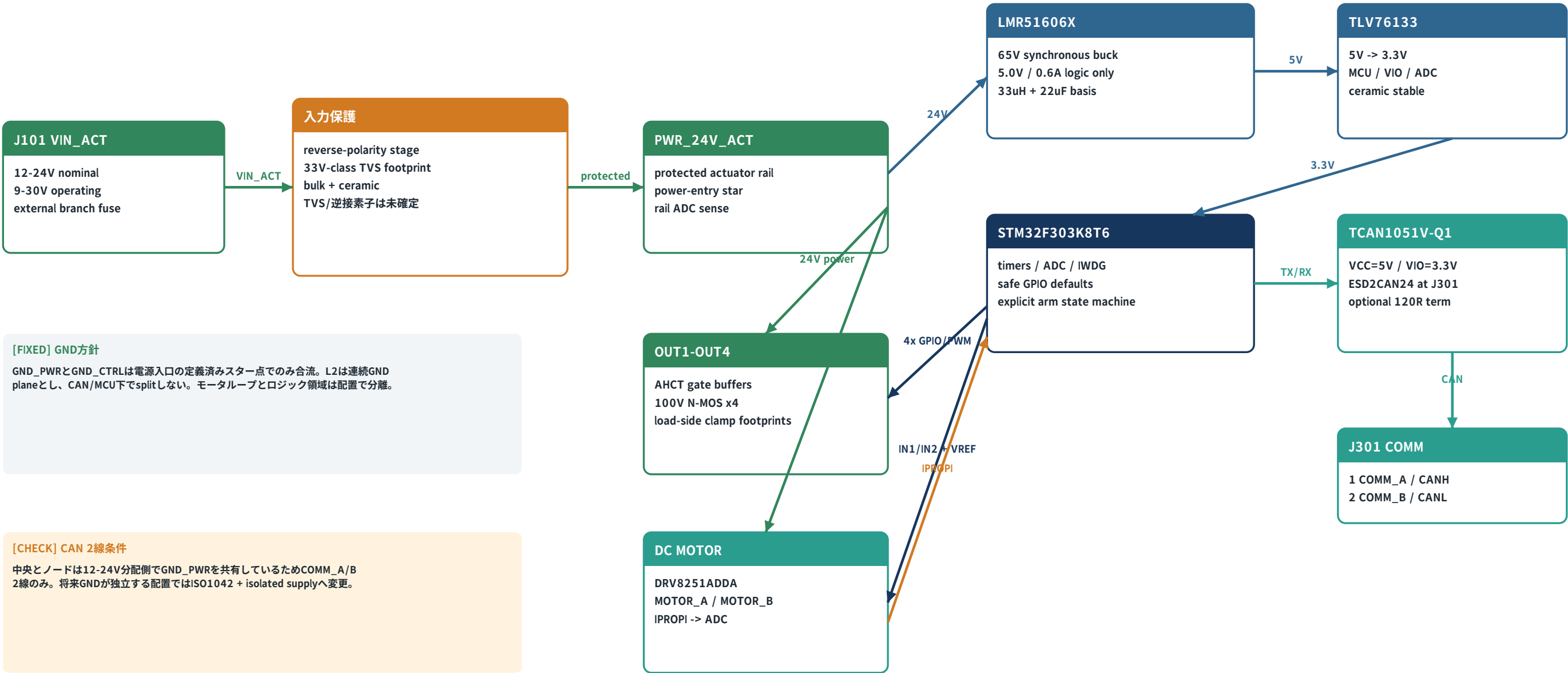


Rev.Aで必ず止める故障

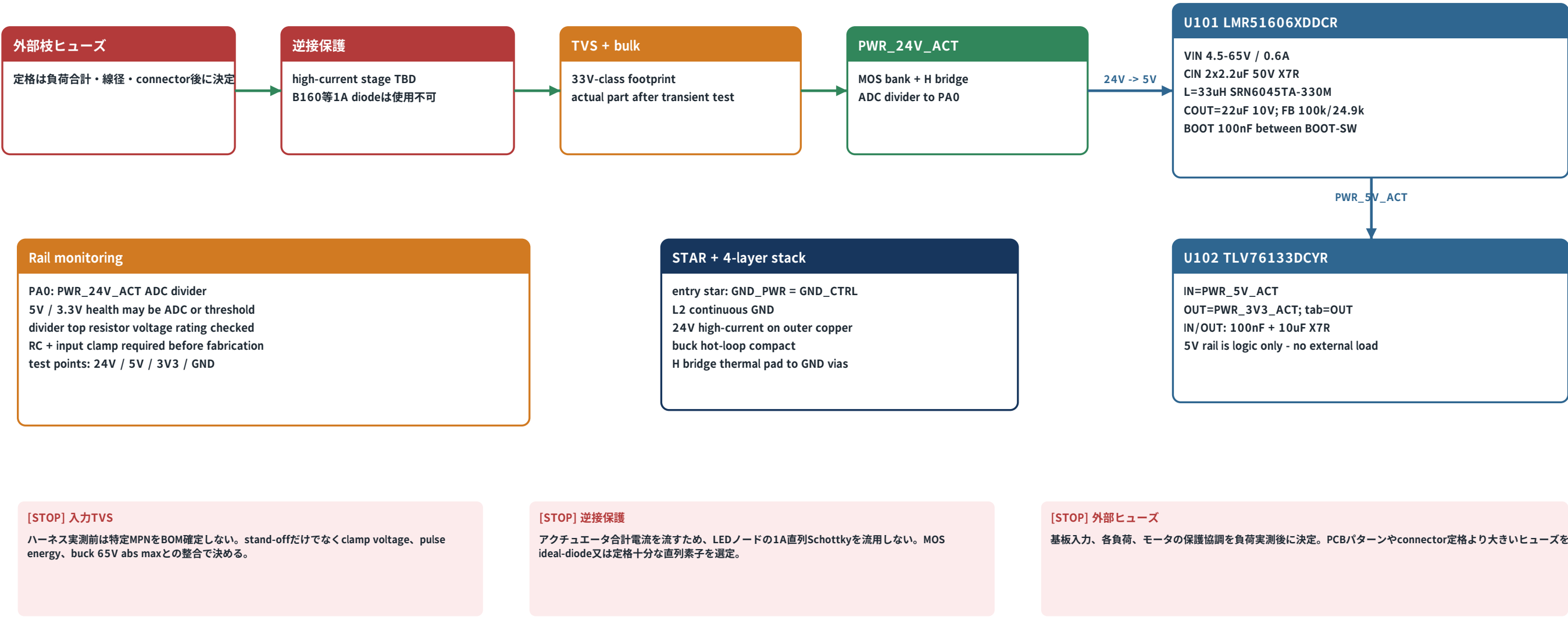


● FIXED ● PROPOSAL ● CHECK ● STOP

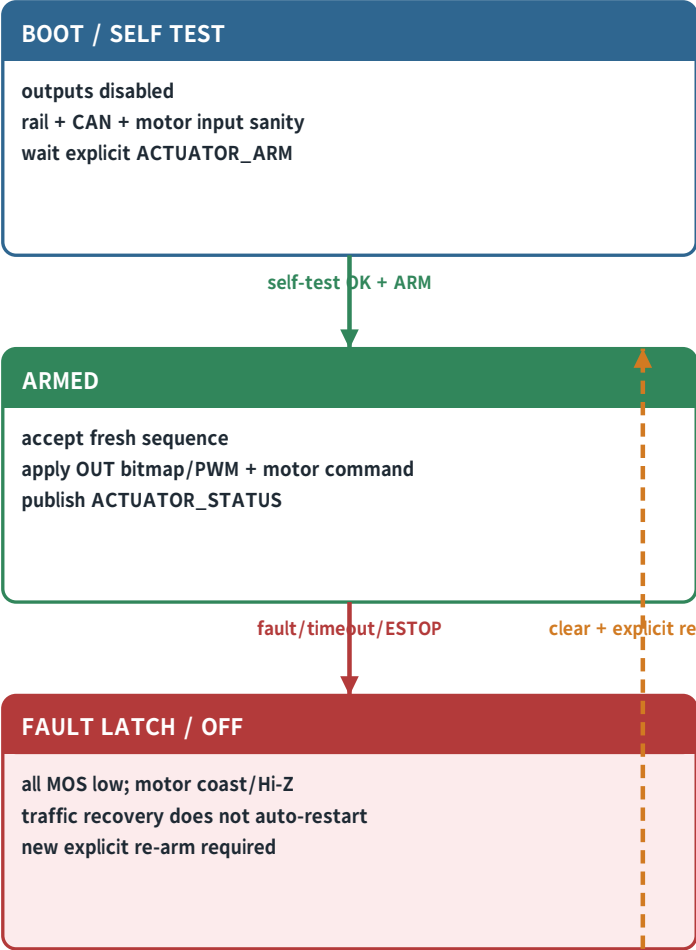
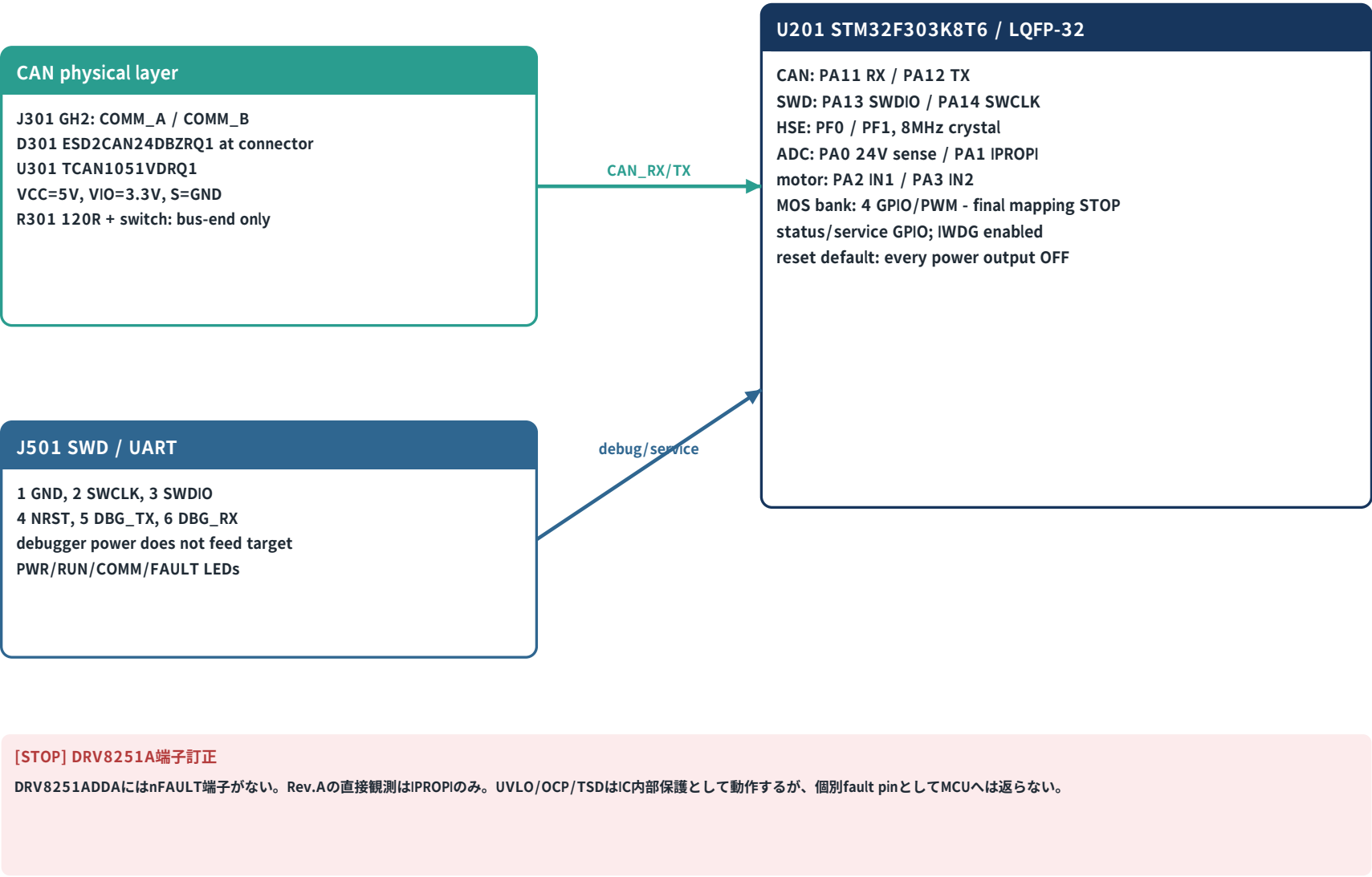
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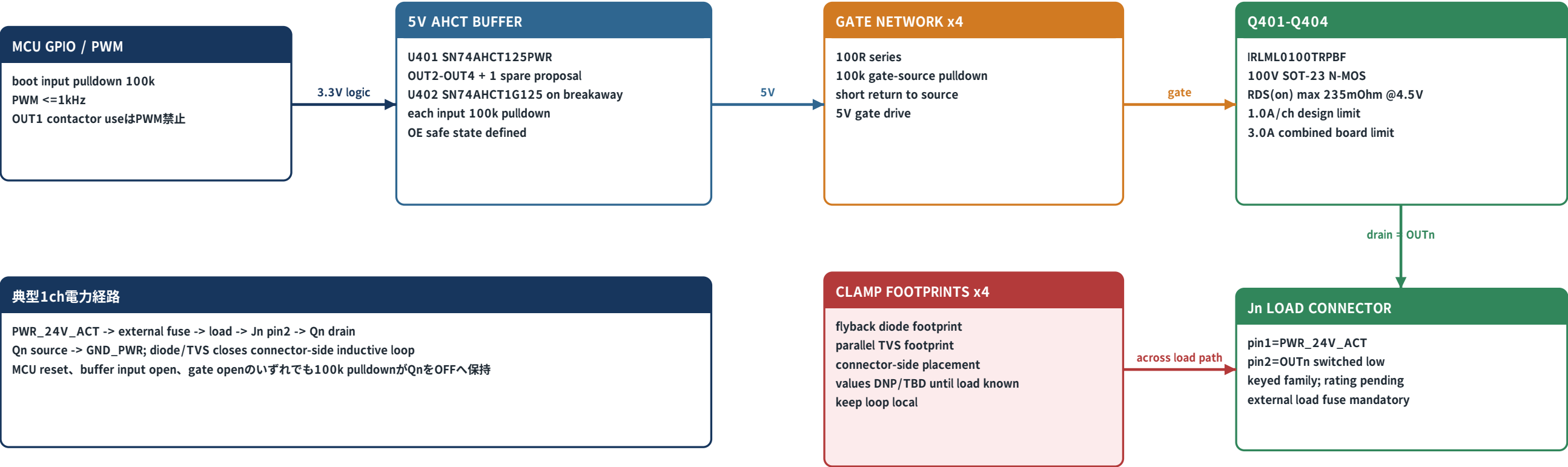


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[CHECK] 熱設計

1A時のMOS損失目安はRDS(on)最大値で約0.235W。銅面・周囲温度・4ch同時率を含め実測する。

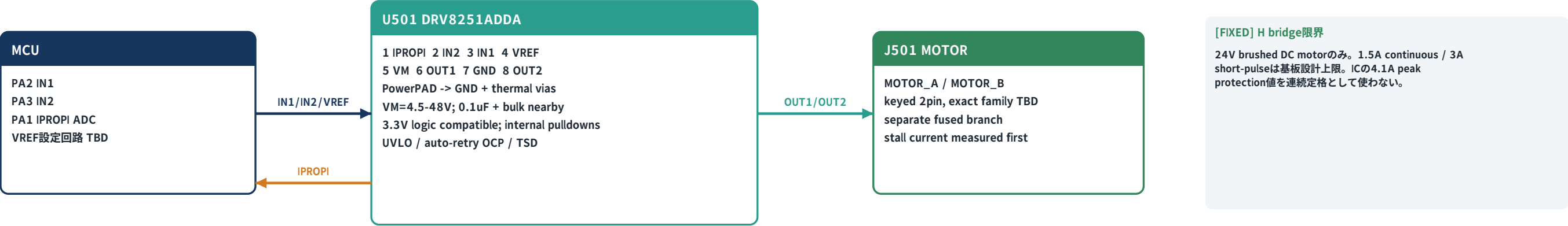
[STOP] 誘導負荷

diodeは低サージだがreleaseが遅い。TVSは速いがVDS stressが増える。負荷ごとに波形を測り、100V margin内で選定。

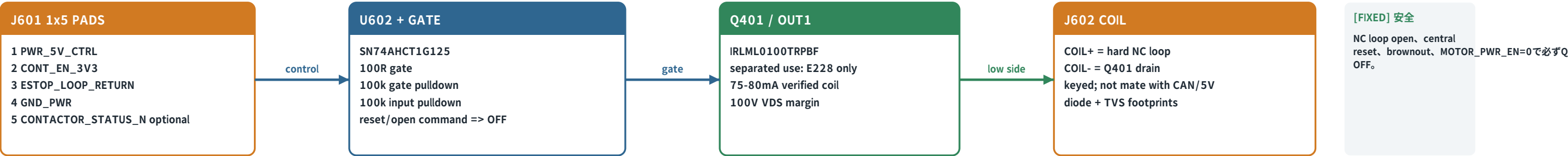
[STOP] GPIO割当

正本内でPA5 contactorとPA6-PB1
OUT1-4が重複。OUT1=breakawayを維持した最終4ch mappingをKiCad前に確定。

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MOUSE-BITE / TAB CUT LINE - 下側ブロックを切離したときだけ中央基板用contactor driver



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Refs	Qty	Function	Manufacturer / MPN or value	Package	Status / note
J101	1	node power input	2-pin keyed power connector, >=8A target	TBD	STOP: exact family/current derating
FEXT	1	external branch fuse	fuse + holder sized after load audit	off-board	STOP: rating pending
Q/D101	1	reverse polarity	high-current MOS ideal-diode or rated series device	TBD	STOP: select by current/transient
D102	1	input surge clamp	33V-class TVS	SMB/SMC TBD	STOP: waveform/energy pending
U101	1	24V to 5V buck	Texas Instruments LMR51606XDDCR	SOT-23-6 DBV	FIXED
L101	1	buck inductor	Bourns SRN6045TA-330M, 33uH	6x6mm SMD	FIXED
C101,C102	2	buck input	2.2uF 50V X7R	1206	FIXED; DC-bias check
C103	1	bootstrap	100nF 16V X7R	0603	FIXED; BOOT-SW
C104	1	buck output	22uF 10V X7R	1206	FIXED; DC-bias check
R101,R102	2	buck feedback	100k / 24.9k, 1%	0603	FIXED; nominal 5.0V
U102	1	5V to 3.3V LDO	Texas Instruments TLV76133DCYR	SOT-223, tab=OUT	FIXED
C111-C114	4	LDO bypass	100nF x2 + 10uF x2, X7R	0603 / 0805	FIXED
U201	1	MCU	ST STM32F303K8T6	LQFP-32	FIXED
Y201	1	8MHz HSE	FC3BAEBDI8.0-T1, CL=8pF	3225 4-pad	FIXED
C201,C202	2	HSE load	10pF COG	0603	FIXED; verify stray C
FB201	1	analog rail bead	Murata BLM18AG601SN1D	0603	FIXED/project common
C2xx	8	MCU/analog decoupling	100nF x4, 1uF x2, 10nF x1, reset 100nF	0603	FIXED group; final refs in KiCad
J501	1	SWD/UART	JST BM06B-GHS-TBT	GH 6-pin vertical	FIXED/project convention
U301	1	CAN transceiver	Texas Instruments TCAN1051VDRQ1	SOIC-8	FIXED
D301	1	CAN ESD	Texas Instruments ESD2CAN24DBZRQ1	SOT-23-3	FIXED
J301	1	CAN connector	JST SM02B-GHS-TB	GH 2-pin SMT	FIXED; 1=A, 2=B
R301,SW301	1+1	optional termination	120R 1% + JS102011SAQN	0603 + slide	DNP unless physical bus end
LED/R	4+4	PWR/RUN/COMM/FAULT	LED 0603 + series resistor	0603	PROPOSAL: color/value at schematic
TP	6	test points	24V, 5V, 3V3, GND, COMM_A, COMM_B	SMD loop/pad	FIXED function

数量は基板1枚あたり。外部ヒューズ・相手側housing/contact・実装予備は含まない。

HUMAN REVIEW ONLY - Rev.A architecture + preliminary BOM - not a fabrication release

Refs	Qty	Function	Manufacturer / MPN or value	Package	Status / note
U401	1	OUT2-OUT4 gate buffer	Texas Instruments SN74AHCT125PWR	TSSOP-14	PROPOSAL: 3 used + 1 spare
U402	1	OUT1/breakaway buffer	Texas Instruments SN74AHCT1G125DCKR	SC70-5	PROPOSAL: stays on tab
Q401-Q404	4	low-side switch	Infineon IRLML0100TRPBF, 100V	SOT-23	FIXED; 1A/ch design limit
RG401-RG404	4	gate series	100R, 1%	0603	FIXED
RGS401-RGS404	4	gate pulldown	100k, 1%	0603	FIXED
RIN401-RIN404	4	buffer input pulldown	100k, 1%	0603	FIXED
C401,C402	2	buffer bypass	100nF X7R	0603	FIXED
D401-D404	4	flyback diode footprints	load-specific diode	TBD footprint	DNP/STOP: load waveform pending
TVS401-TVS404	4	fast release clamps	load-specific TVS	TBD footprint	DNP/STOP: Vclamp pending
J401-J404	4	24V load outputs	keyed 2-pin >=1A/channel	TBD	STOP: exact connector/fuse
U501	1	brushed motor H bridge	Texas Instruments DRV8251ADDA	HSOP-8 PowerPAD	FIXED; no nFAULT pin
C501	1	VM ceramic bypass	100nF >=50V X7R	0603/0805	FIXED; at VM-GND
C502	1	motor local bulk	low-ESR electrolytic/polymer >=50V	TBD	STOP: capacitance/ripple pending
RIPROPI	1	current mirror resistor	value set from current range/ADC	0603	STOP: calculate from DS
VREF net	2-3	current regulation threshold	divider/DAC/filter as selected	0603	STOP: topology/value pending
J501M	1	motor output	keyed 2-pin >=1.5A cont / 3A pulse	TBD	STOP: exact family/thermal
J601A/B	2	breakaway interface pads	2.54mm 1x5 unpopulated through-hole	THT pads	FIXED; normally no header
J602	1	E228 coil connector	keyed 2-pin, incompatible with CAN/5V	TBD	STOP: exact family
D601	1	E228 safe flyback	diode sized for 24V/80mA coil	TBD	PROPOSAL for first test
TVS601	1	E228 fast release	parallel TVS footprint	TBD	DNP/STOP: waveform pending
RADC/CADC	3-5	24V/IPROPI ADC conditioning	divider + RC + clamp network	0603	STOP: exact values/protection
PCB	1	board	4-layer, outer 1oz minimum, L2 solid GND	mouse-bite edge tab	FIXED architecture

[CHECK] BOMの意味

FIXEDは採用品、PROPOSALは回路分割案、DNP/STOPはfootprint候補を置いても実装・発注確定しない。価格・在庫は未調査。

[STOP] 不足している設計値

入力逆接/TVS、各負荷clamp、motor bulk、RIPROPI/VREF、全power connector/fuseは実負荷とハーネス測定後に確定する。

KiCad着手前に閉じるSTOP項目

- ☐
1.
- DRV8251AにnFAULT pinはない。status/protocol/firmwareのfault表現をIPROPI + inferred protectionへ修正するか、nFAULT必須ならdriverを再選定。
- ☐
2.
- OUT1=breakawayという境界と、PA5 contactor / PA6-PA7-PB0-PB1 OUT1-4というGPIO割当重複を解消。4 MOS総数は維持。
- ☐
3.
- VIN_ACT入力connector、reverse-polarity素子、外部branch fuseを最大同時負荷・wire gauge・温度で選定。
- ☐
4.
- input TVSは実ハーネスtransientを測定し、33V stand-off / clamp energy / 65V buck limitで確定。
- ☐
5.
- 4出力それぞれの実負荷を確定し、flyback diode / TVS / fuse / connectorを個別選定。
- ☐
6.
- 使用DC motorのstall currentを測り、RIPROPI、VREF、bulk capacitor、connectorを計算。
- ☐
7.
- CAN topologyを確認し、120R terminationは物理bus endだけ実装。
- ☐
8.
- mouse-bite加工ルール、tab copper bridge、切離し後clearanceを基板業者仕様で確認。

Bring-up順序

- 1
- 5V current-limited input: 3.3V, reset defaults, SWD, CAN, watchdog, every gate OFF.
- 2
- 24V no-load: input transient/rail ADC, buck temperature, DRV8251 state, CAN timeout.
- 3
- Resistive load: 0.25A -> 0.5A -> 1.0A/ch, single then combined, record MOS/connector/copper temperature.
- 4
- Inductive loads: capture Q VDS and release time; populate diode/TVS only from measured requirement.
- 5
- Motor: measured stall current, current-limited source/fuse, direction/PWM/coast/brake, IPROPI and thermal test.
- 6
- Separated contactor tab: dummy 24V/80mA then E228; open each NC, reset/brownout, record release waveform.
- 7
- KiCad ERC=0, PCB DRC=0, netlist-to-pin-table review, then fabrication release.

Primary data sheets checked

- ST DS9866 Rev.8 - STM32F303x6/x8; CAN 2.0B, LQFP-32 pin functions.
- TI LMR51606 SLUSEY1B - 4.5-65V, 0.6A; 400kHz X variant application values.
- TI DRV8251A SLVSFU6 - 4.5-48V, 4.1A peak, HSOP-8 pinout, IPROPI; no nFAULT pin.
- TI TCAN1051-Q1 Rev.D / ESD2CAN24-Q1 Rev.D / TLV761 Rev.D / SN74AHCT125 Rev.R.
- Infineon IRLML0100 - 100V SOT-23; RDS(on) max 235mOhm at VGS=4.5V.